

Use of Semaphores

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Introduction

The Hungry Birds Problem

Implementation

To verify the output of the code and if it runs correctly, we implemented a counter so that the parent bird refills the plate with worms maximum 3 times. However, the code can be modified to run indefinitely. In addition, we implemented a signal mechanism so that when the plate is empty only one baby bird signals to the parent that the plate is empty, this ensures that the plate is refilled only once when its empty. Semaphores were used for:

- Mutual exclusion lock which ensures only one bird accessing the plate at a time.
- Plate empty semaphore which allows baby bird notifying the parent when the plate is empty. full semaphore which allows the parent bird notifying the baby birds the food is refilled.

Results

A sample output is presented below

```
baby bird 0 ate a worm. 9 worms left  
baby bird 2 ate a worm. 8 worms left  
baby bird 1 ate a worm. 7 worms left  
baby bird 0 ate a worm. 6 worms left  
baby bird 1 ate a worm. 5 worms left
```

```
baby bird 2 ate a worm. 4 worms left
baby bird 2 ate a worm. 3 worms left
baby bird 1 ate a worm. 2 worms left
baby bird 0 ate a worm. 1 worms left
baby bird 0 ate a worm. 0 worms left
Baby bird 1 chirping, The plate is empty
Baby bird 2 chirping, The plate is empty
Parent birds gets 6 worms.
baby bird 0 ate a worm. 5 worms left
baby bird 2 ate a worm. 4 worms left
baby bird 1 ate a worm. 3 worms left
baby bird 1 ate a worm. 2 worms left
baby bird 2 ate a worm. 1 worms left
baby bird 0 ate a worm. 0 worms left
Parent birds gets 7 worms.
```

Conclusion

This solution can cause some baby birds to receive more food than others, since all birds awaken when the plate is full. As seen in the output above, some baby birds receive more food than others.

The Bear and Honeybees Problem

Implementation

In the assignment we used two semaphores for synchronization:

- Full semaphore that signals the bear when the honeypot is full.
- Empty semaphore that controls the bees that add honey to the honeypot.

The bees need to wait for permission before adding honey to the honeypot. If empty is 0, the bees need to wait until the bear eats and resets the honeypot. When the honeypot reaches the capacity, the last bee wakes up the bear to eat all the honey and empty the honeypot. After the bear has emptied the honeypot, it signals the bees using empty semaphore to start filling the pot again.

Conclusion

All bees will get the chance to add honey to the honeypot. Only one bee wakes up the bear ensuring the bear is not woken multiple times. A queue system can be used for more fair order of bees adding honey to the honeypot.